

# **GEOGRAPHY**

# **PHYSICAL GEOGRAPHY**



## **CLIMATOLOGY**

**MODULE – 21**

**JET STREAMS**

# JET STREAM

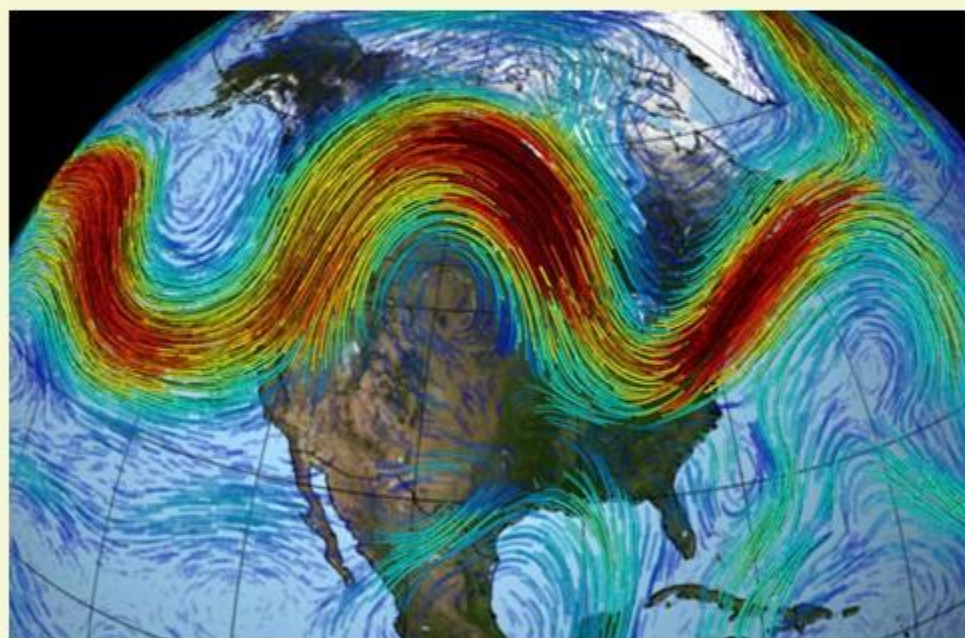


## WHAT DO YOU MEAN BY JET STREAM?

- 1 The strong and rapidly moving circumpolar westerly air circulation in a narrow belt of few hundred kilometres width in the upper limit of troposphere is called jet stream.
- 2 Jet stream was discovered during the Second World War when American jet bomber fighter planes while flying towards Japan (from east to west) found obstructions of an air circulation which was moving in opposite direction (west to east) resulting into marked reduction in the velocity of jet fighter planes. These planes registered marked increase in their velocity while they used to return to their bases.



# JET STREAM



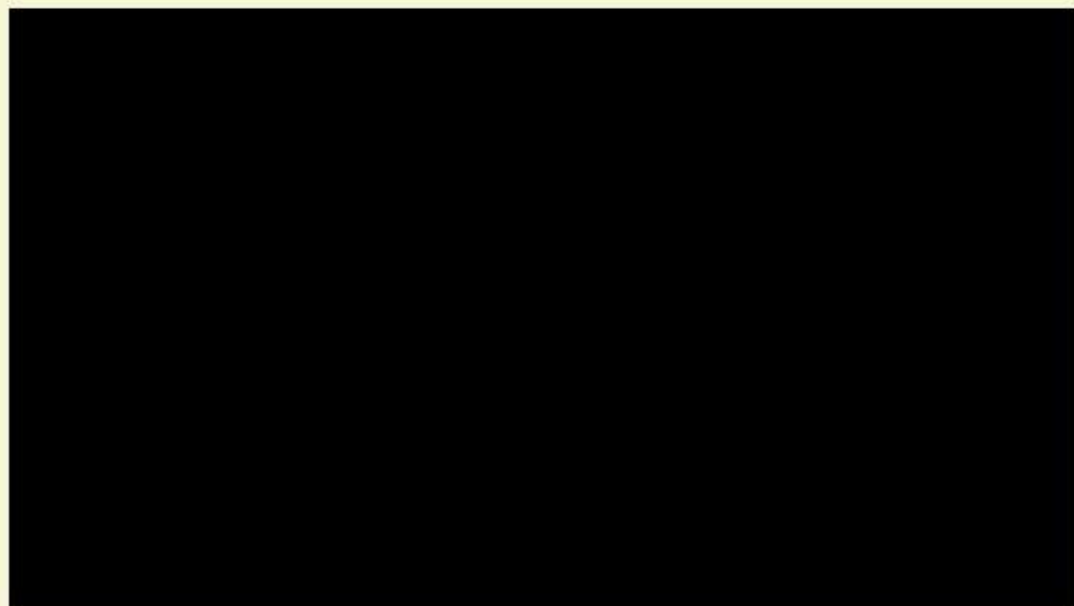
## WHAT DO YOU MEAN BY JET STREAM?

**3** After careful study of this phenomena, it was found that there was a strong upper air circulation from west to east in the upper portion of troposphere which presented obstruction in the free movement of jet fighter planes.

**4** Based on this fact, **westerly strong meandering upper air circulation** was called as jet stream.

# JET STREAM

## PROPERTIES OF JETSTREAM



**Credits: NASA**

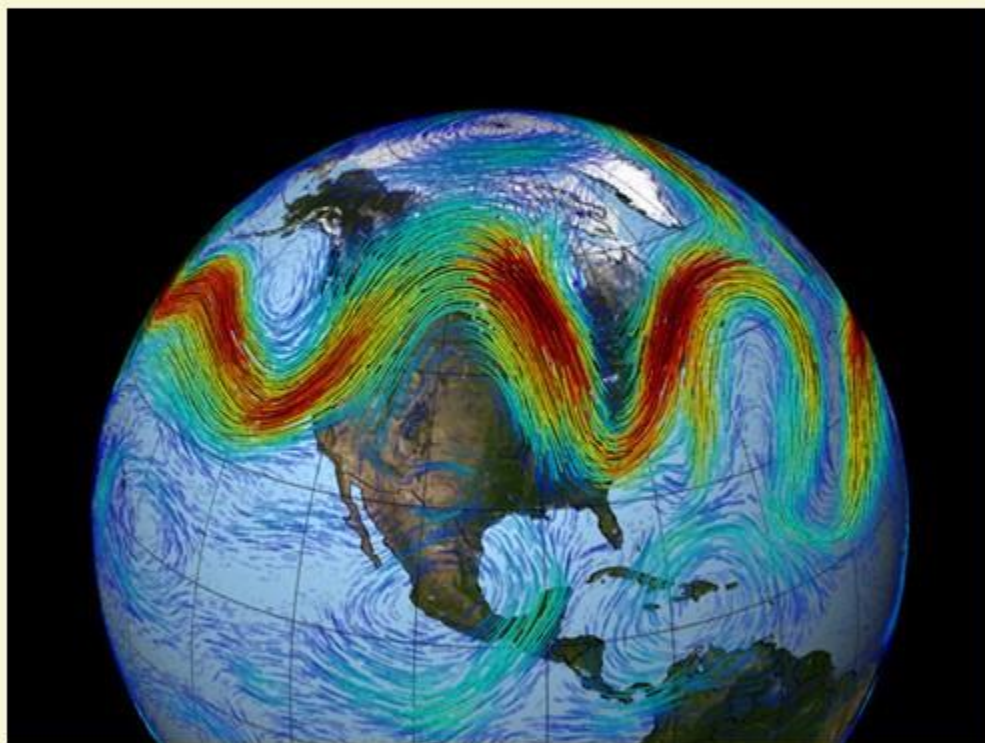
Shows a winding polar jet stream. Here, the highest wind velocity is colored red; slower winds are blue.

- 1 The circulation of jet streams is from the **west to east** in a narrow belt of a few hundred kilometres width at the height of **7.5 -14 km in the upper troposphere**.
- 2 On an average, jet streams measure thousands of kilometres in length, a few hundred kilometres in width and a few kilometres (2-4 km) depth.
- 3 Generally, their circulation is observed between **poles and 20<sup>0</sup> latitudes** in both the hemispheres.
- 4 These are also called **circum-polar whirl** because these move around the poles in both the hemispheres.



# JET STREAM

## PROPERTIES OF JETSTREAM

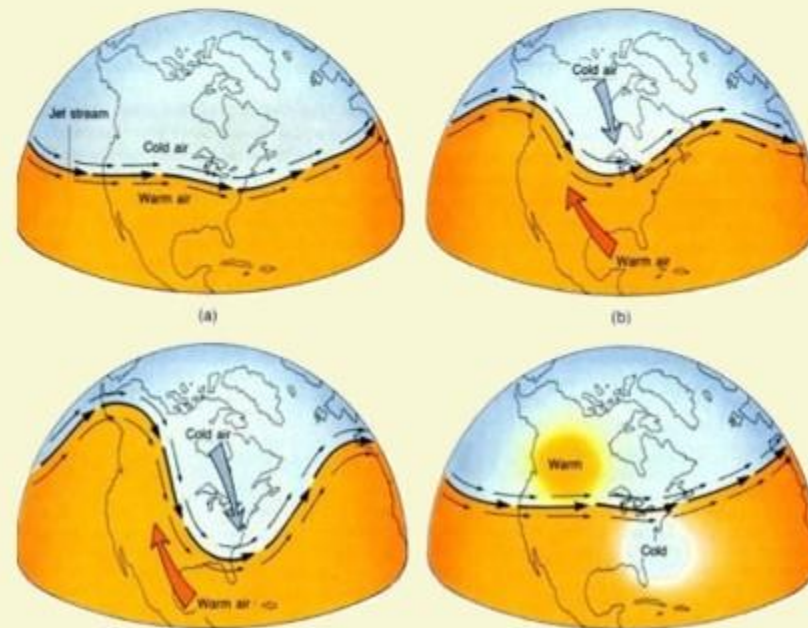
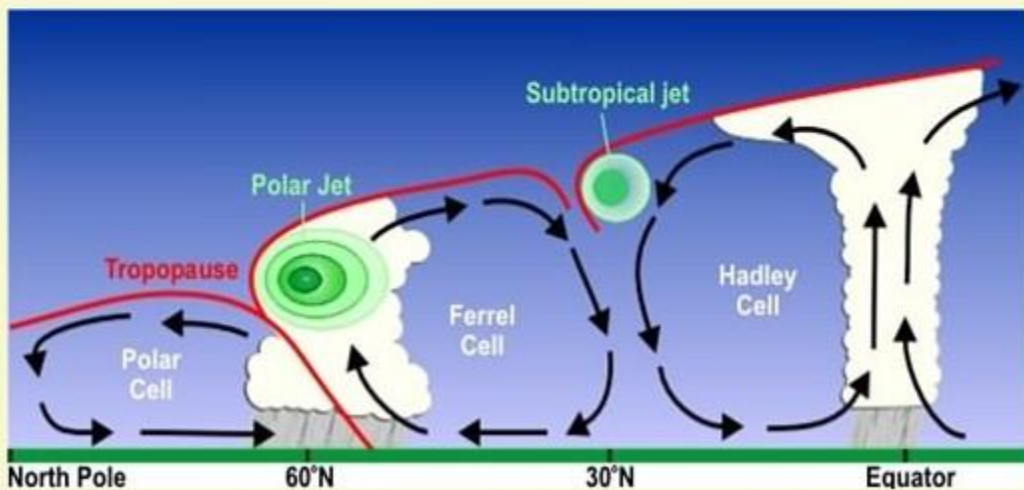


5 The **vertical wind shear** of jet streams is 18-36 kmph meaning thereby the **wind velocity above or below jet stream decreases by 18-36 kmph**.

6 The circulation path (trajectory) is **wavy and meandering**. When the Jetstream attains the meandering stage, they are known as **Rossby waves**.

7 Because of the meandering effect, there is displacement of polar cold air masses towards equator and of tropical warm air masses towards the poles.

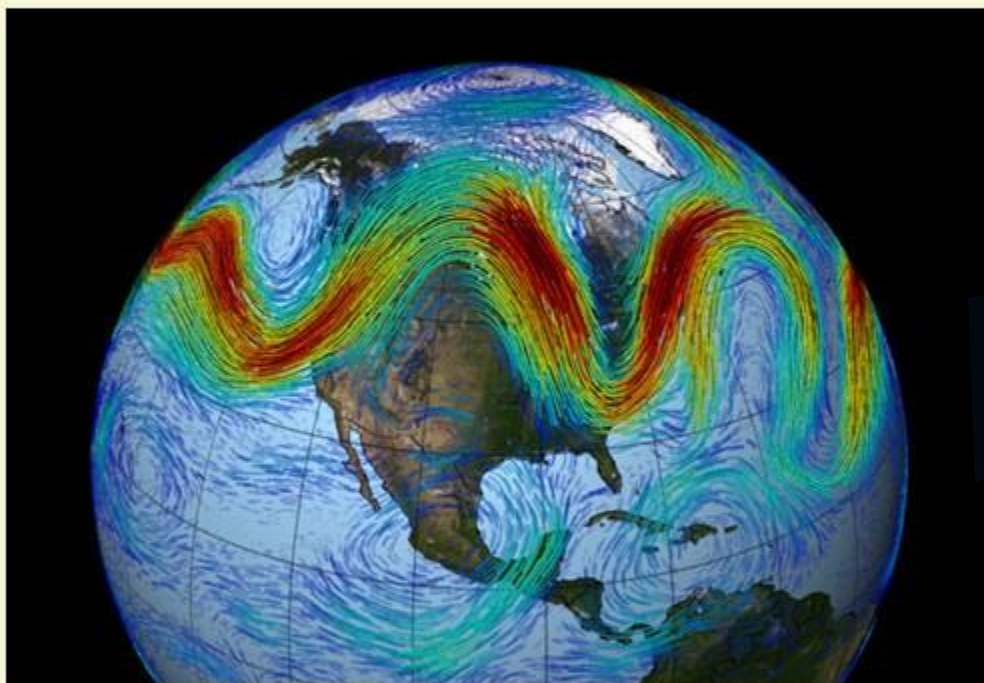
# PROPERTIES OF JETSTREAM





# JET STREAM

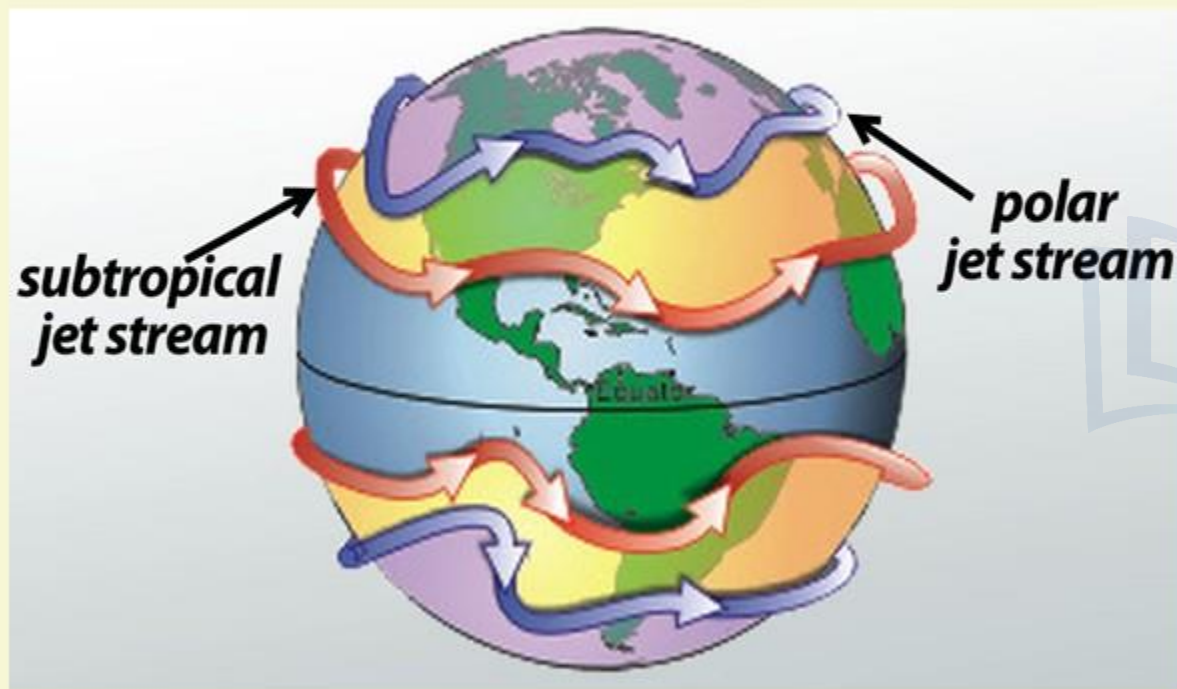
## PROPERTIES OF JETSTREAM



- 7 There is seasonal change in the wind velocity in jet streams wherein these becomes strong during winter season and the wind velocity becomes twice the velocity of the summer season.
- 8 The extent of jet streams narrows down during summer season because of their northward shifting while these extend up to  $20^{\circ}$  latitudes during winter season.

# JET STREAM

## TYPES OF JET STREAMS



### 1. POLAR FRONT JET STREAMS

- 1 Polar front jet streams are **formed** above the **convergence zone** (**40-60° latitude**) of the surface polar cold air mass and sub-tropical warm air mass.
- 2 The thermal gradient is **steepened** because of convergence of two **contrasting air masses**. These move from west towards the east **but are irregular** in comparison to the subtropical westerly jetstream.



# JET STREAM

## TYPES OF JET STREAMS

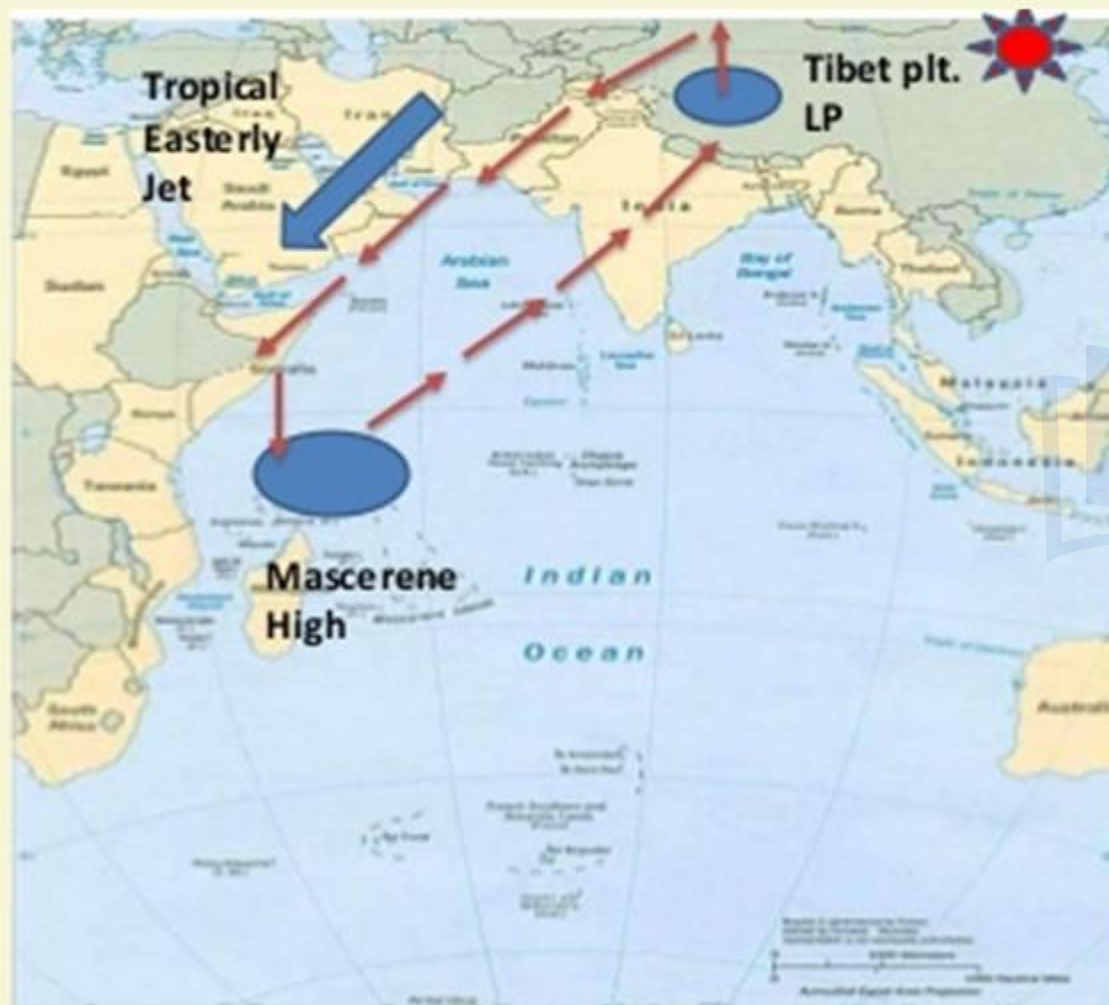


### 2. SUBTROPICAL WESTERLY JET STREAMS

- 1 Subtropical westerly jet streams move in the upper troposphere to the north of subtropical surface high pressure belt (at the poleward limit of the Hadley cell in both the hemispheres) i.e above  $30^{\circ}$ - $35^{\circ}$  latitudes.
- 2 Their circulation is from the west to east in more regular manner than the polar front jet streams

# JET STREAM

## TYPES OF JET STREAMS



### 3. TROPICAL EASTERLY JET STREAMS

Tropical easterly jet streams develop in the upper troposphere above surface easterly trade winds over India and Africa during summer season due to intense heating of Tibetan plateau and play important role in the mechanism of Indian monsoon



# JETSTREAMS

## UPSC 2008 GS-1 QUESTION

**The winter rains in North India are largely related to Jet Streams and Western Disturbances. Bring out the relationship. (UPSC 2008)**

### HINTS

- 1 Subtropical Westerly Jetstream.
- 2 Cyclonic rainfall from Mediterranean brought towards North India aka Western Disturbances.
- 3 STWJ over Northern Plains; Retreating Monsoon Season.
- 4 Good rainfall over Tamilnadu, Andhra Pradesh because of North-East monsoon during the winter season.

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